Rigorous Generalization of a Nested Radical and Gamma Function Limit

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1 Introduction and Statement of the Theorem

We seek to evaluate and generalize the following limit:

$$\mathcal{L} = \lim_{x \to 0} \frac{\left(\sqrt{a - bx\sqrt{a - bx\sqrt{\cdots}}}\right)\left(\tan\left(\frac{\pi}{2} - cx\right)\right)}{\Gamma(kx)} \tag{1}$$

where $a, c, k \in \mathbb{R}^+$ and $b \in \mathbb{R}$.

Theorem 1. *For $a, c, k > 0$ and $b \in \mathbb{R}$, the limit evaluates to:*

$$\mathcal{L} = \sqrt{a}\left(\frac{k}{c}\right) \tag{2}$$

2 Proof of the Theorem

To prove the theorem, we decompose the expression into its three constituent asymptotic components as $x \to 0$.

Lemma 2 (The Nested Radical)**.** *Let $R(x) = \sqrt{a - bx\sqrt{a - bx\sqrt{\cdots}}}$. For sufficiently small x, $\lim_{x \to 0} R(x) = \sqrt{a}$.*

Proof. Formally, $R(x)$ is defined as the limit of the sequence $y_0 = \sqrt{a}$, $y_{n+1} = \sqrt{a - bxy_n}$. Assuming convergence for a chosen domain of x near zero, the infinite nested radical satisfies the quadratic relation:

$$[R(x)]^2 = a - bxR(x) \implies [R(x)]^2 + bxR(x) - a = 0 \tag{3}$$

Applying the quadratic formula and selecting the positive root (as $R(x)$ represents a principal square root bounded below by zero for valid domains), we obtain the closed-form expression:

$$R(x) = \frac{-bx + \sqrt{b^2x^2 + 4a}}{2} \tag{4}$$

Taking the limit as $x \to 0$, we find:

$$\lim_{x \to 0} R(x) = \frac{\sqrt{4a}}{2} = \sqrt{a} \tag{5}$$

□

Lemma 3 (The Trigonometric and Gamma Asymptotics)**.** *As $x \to 0$, the ratio of the trigonometric factor to the Gamma function factor converges to $\frac{k}{c}$.*

Proof. By the complementary angle identity, $\tan\left(\frac{\pi}{2} - cx\right) = \cot(cx)$. Using the small-angle approximation for the cotangent function, we have:

$$\cot(cx) = \frac{\cos(cx)}{\sin(cx)} \sim \frac{1}{cx} \quad \text{as} \quad x \to 0 \tag{6}$$

For the denominator, we utilize the fundamental property of the Gamma function, $\Gamma(z + 1) = z\Gamma(z)$. Thus:

$$\Gamma(kx) = \frac{\Gamma(kx + 1)}{kx} \tag{7}$$

Since $\Gamma(1) = 1$, the asymptotic behavior near zero is:

$$\Gamma(kx) \sim \frac{1}{kx} \quad \text{as} \quad x \to 0 \tag{8}$$

Evaluating the ratio of these limits yields:

$$\lim_{x \to 0} \frac{\tan\left(\frac{\pi}{2} - cx\right)}{\Gamma(kx)} = \lim_{x \to 0} \frac{\frac{1}{cx}}{\frac{\Gamma(kx+1)}{kx}} = \lim_{x \to 0} \left(\frac{kx}{cx} \cdot \frac{1}{\Gamma(kx + 1)} \right) = \frac{k}{c} \tag{9}$$

□

Conclusion of the Main Theorem:
Since the limits of the individual factors exist and are finite as $x \to 0$, we may apply the product rule for limits:

$$\begin{aligned}
\mathcal{L} &= \left(\lim_{x\to 0} R(x)\right) \cdot \left(\lim_{x\to 0} \frac{\tan\left(\frac{\pi}{2} - cx\right)}{\Gamma(kx)}\right) \\
&= \left(\sqrt{a}\right) \cdot \left(\frac{k}{c}\right) \\
&= \sqrt{a}\left(\frac{k}{c}\right)
\end{aligned}$$

This concludes the proof. $\square$